RV College of Engineering®



OptiProductio

Arduino-Based Real-Time WIP Monitoring and Bottleneck Detection

INDUSTRIAL ENGINEERING & MANAGEMENT

Go, change the world

RV College of Engineering

Outline of Tentative Steps

OptiProductio: Arduino-Based Real-Time WIP Monitoring and Bottleneck Detection

Phase I 29 Oct 2026	Phase II __ / __ / 2026	Phase III __ / __ / 2027
• Pick and define the problem • Literature review • SMART objectives (PO1 to PO4) • Design the 3-station line and parts list • Plan the method and timeline	• Wire sensors to Arduino UNO • Python logger and dashboard • Check counts against a hand tally • First runs and partial results • Dashboard on Bosch public data	• Rebalance and WIP-limit experiments • Results and discussion • Live prototype demo • Report, poster and PPT • IEEE-format paper draft

Status today: Phase I planning done. Hardware build in progress. Red blanks = dates not announced yet.

RV College of Engineering

EL Phase-I Presentation Outline

- Introduction
- Literature Review
- Problem Definition
- Objectives
- Methodology and System Pipeline
- Measurement Parameters
- Research Process
- Proposed Outcomes
- Timeline of the Project
- Conclusion
- References
- Expected Outcome by the End of First Year

RV College of Engineering

INTRODUCTION

OVERVIEW OF THE TOPIC:

OptiProductio: a low-cost system that watches a small production line in real time and shows how well it is working.

How: sensors on a 3-station model line send every event (part done, part rejected, line stopped) to an Arduino UNO, which passes it to a Python dashboard on a laptop.

Why: small factories often record output by hand, so delays, pile-ups and the slowest station stay hidden until the end of the shift [2], [4].

What it adds: an alert when too many parts pile up inside the line (a WIP limit), based on the CONWIP idea [5].

Production count	**Cycle time**	**Inventory**	**Workflow**
IR sensor at each station exit	time each station takes per part	raw material bin level and WIP	bottleneck, stops and pile-ups

Project title: Arduino-Based Real-Time WIP Monitoring and Bottleneck Detection. EL topic: Smart Production Workflow and Resource Optimization System. Deliverable: production efficiency dashboard.

RV College of Engineering

Literature Review

Key studies and findings

Study	What they did	Key finding
Hwang et al., 2017 [2]	IoT performance-measurement model built on the ISA-95 and ISO 22400 standards	IoT timestamp data can give OEE numbers in real time
Ng Corrales et al., 2020 [3]	Systematic review of OEE: 862 articles screened down to 186	Many industries adapt OEE, because the basic formula alone is not enough
Wicaksono et al., 2024 [4]	Replaced manual OEE logging with an ESP32 board over MQTT	Works in real time, but reports up to 16% OEE error
Spearman et al., 1990 [5]	Proposed CONWIP: a fixed cap on the total parts inside a line	A simple pull alternative to kanban that limits WIP
Thürer et al., 2023 [6]	Tested kanban, CONWIP and a hybrid on a physical lab-scale line	WIP limits cut the time a job spends in the line but can lower throughput

RV College of Engineering

Literature Review (contd.)

- **Trends in literature:** low-cost IoT boards now give real-time OEE [2], [4]; WIP limits like CONWIP are mostly tested in simulation or lab models [6], [9]; public line data such as the Bosch dataset is used mainly to predict failed parts [8].
- **Theoretical framework:**
  - **Little's law:** average WIP = throughput × average flow time [7]. Cut WIP at the same throughput and parts spend less time in the line.
  - **OEE = Availability × Performance × Quality** [3], [10].
  - **CONWIP:** release a new part only when a finished part leaves [5].
- **Relevance to our project:** monitoring papers report numbers but do not control the flow, and WIP-limit papers control the flow but are rarely tied to a live, measured line [6], [9]. Low-cost builds rarely check their accuracy against a manual count [4]. OptiProductio does all three on one small line.

RV College of Engineering

Problem Definition

- **Clear statement of the problem:** on a small manual line, work does not flow evenly. Four bottlenecks keep appearing, and today they are spotted only when someone walks past or from a paper tally at the end of the shift:
  - **Slow station:** one task takes longer (our Station 2), so parts pile up in front of it and Station 3 stands idle.
  - **Breakdown:** a stopped station blocks the stations before it and starves the ones after it.
  - **Empty raw-material bin:** Station 1 runs out of parts without warning and the whole line starves.
  - **Burst of rejects:** bad parts load the inspection station, so good output drops while the line looks busy.
- **Pile-ups:** if parts keep being released, WIP keeps growing in front of the slowest station [11] and every part waits longer [7].
- **Context:** MSMEs gave 30.1% of India's GDP (2022-23) and 45.73% of exports (2023-24) [1]. Manual OEE recording is slow [4], and ordinary systems struggle to capture what is happening on the floor in real time [2].
- **Gap and novelty:** low-cost monitoring and WIP control are studied separately [4], [9]. OptiProductio puts both on one prototype (under ₹2,000, our estimate) and checks its counts against a hand tally.

RV College of Engineering

Objectives

- **Main objective:** build a low-cost prototype that measures production count, cycle time, inventory and WIP on a 3-station model line in real time, and uses them to show which of four bottlenecks is holding the line up: a slow station, a breakdown, an empty raw-material bin or a burst of rejects. Then test a WIP limit that stops parts piling up in front of it.

PO	Specific objective	Measurable target	By
PO1	Count parts at all 3 stations with IR sensors	Count error ≤ 2% against a hand tally, over three 10-minute runs	Phase II
PO2	Live dashboard: count, cycle time per station, WIP, bin level, downtime, OEE proxy	Refreshes every 2 s or faster; all 6 numbers on one screen	Phase II
PO3	Name the bottleneck, then rebalance the slow station	Right cause shown in 4 of 4 staged tests (slow S2, breakdown, empty bin, rejects); throughput and flow time before and after, 3 runs each	Phase III
PO4	WIP-limit alert (LED, buzzer, dashboard)	Alert fires at the cap every time; trade-off reported for caps 2, 4, 6	Phase III

Why achievable: parts are standard IDEA Lab kit, the code is already written and tested on sample data, and each PO is checked by a simple count or timer. **Expected contribution:** a cheap, open template for monitoring and WIP control that small units and student labs can copy.

RV College of Engineering

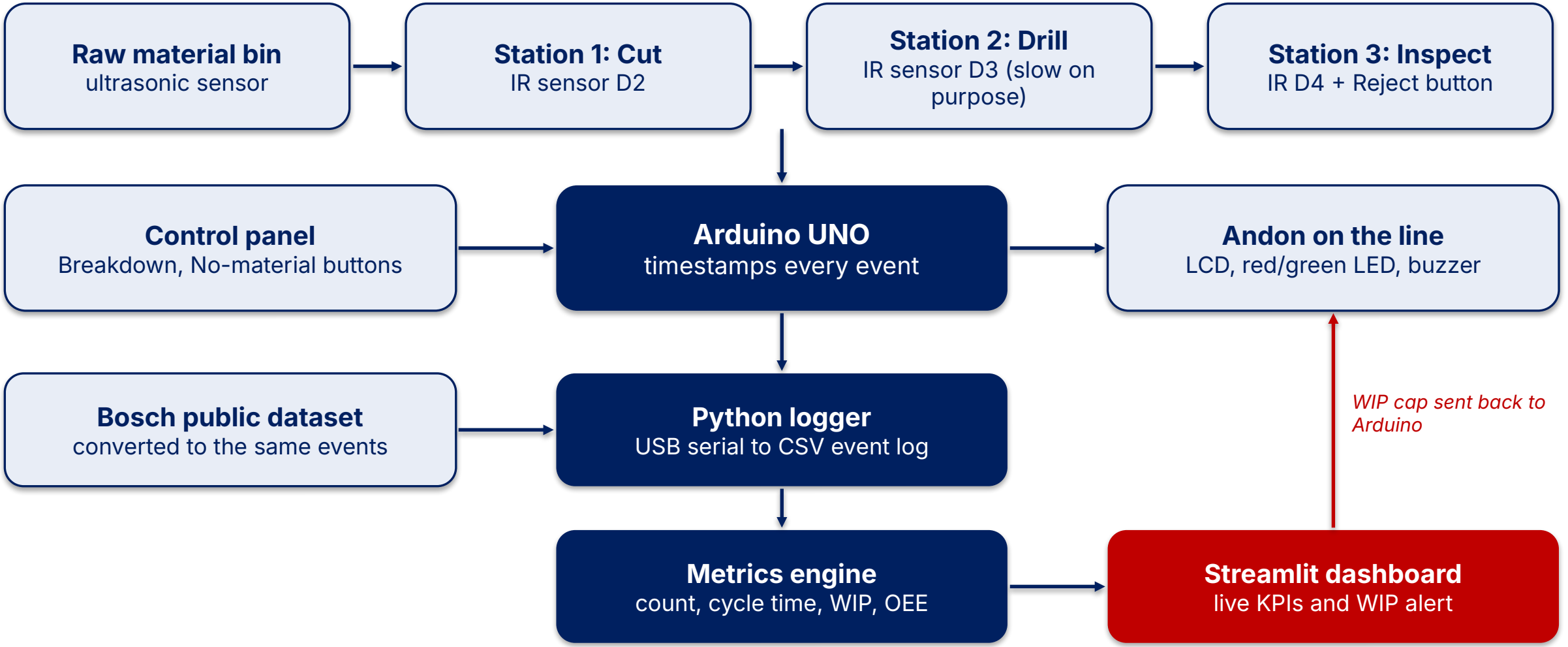
Methodology

- **Design:** quantitative and experimental. Build the line, measure it, change one thing at a time (rebalance, then WIP limit) and compare the numbers.
- **Data collection:** IR sensors time every part leaving each station; buttons log breakdown, no-material and reject; an ultrasonic sensor reads the raw material bin. The Arduino stamps each event in milliseconds and sends it over USB to a Python logger (CSV). Secondary data: the public Bosch production line dataset [8], [12].
- **Data analysis:** throughput, cycle time per station, WIP and flow time from the event log; Little's law as a cross-check [7]; OEE proxy = A × P × Q [3]; mean and spread across 3 repeat runs.
- **Tools:** Arduino IDE (C++), Python (pyserial, pandas, Streamlit), Excel for result tables.
- **Risks and fallbacks:** miscounts (cardboard gates, debounce, hand tally check); hardware failure on demo day (replay a saved run); Bosch time units are not published (use only for relative comparison).

RV College of Engineering

System Pipeline

Physical line (parts moved by hand)



RV College of Engineering

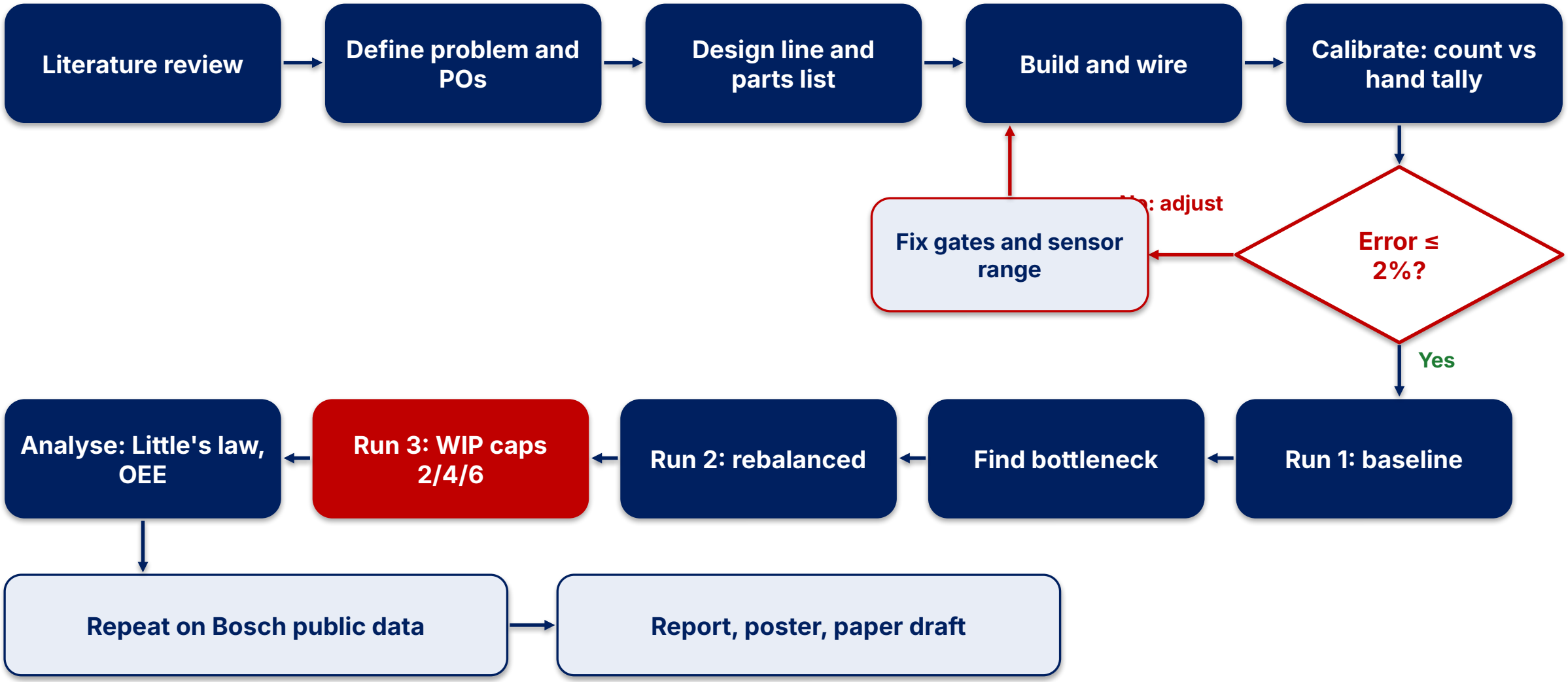
Measurement Parameters

No machine-learning model is trained in this project. These are the measurement settings we will report.

Item	Value
Data sources	Live line event log (CSV) and a Bosch sample: first 50,000 rows, line L3, 3 busiest stations [12]
Event format	time (s), event (PASS / SCRAP / DOWN / UP), station, reason
Sensor timing	IR debounce 400 ms; button debounce 200 ms; status sent every 1 s; dashboard refresh 2 s
Cycle time (per station)	finish(k) − max(arrive(k), finish(k−1)), so stations after the bottleneck do not look slow
WIP	parts released at Station 1 minus parts that left Station 3 (good or scrap)
Throughput / flow time	good parts per minute / exit time − entry time per part; checked with Little's law [7]
OEE proxy	A × P × Q [3]; ideal cycle time set by the team: **__ s**
Experiments	baseline, rebalanced, WIP cap 2 / 4 / 6; 10 minutes per run, 3 repeats each
Targets	count error ≤ 2%; Little's law gap ≤ 10%; alert fires every time WIP reaches the cap

RV College of Engineering

Research Process



RV College of Engineering

Proposed Outcomes

- **Expected results:** a working prototype (3-station line and live dashboard) and a results table comparing baseline, rebalanced and WIP-capped runs: throughput, average WIP, flow time, cycle time per station, OEE proxy and scrap rate.
- **Success criteria:** PO1 to PO4 met. Improvement after rebalancing: __ % (to be measured, not assumed).
- **Potential impact:** shows that count, cycle time, inventory and workflow can be watched live with low-cost parts, as a template for small units and student labs.
- **Limits we will state:** OEE is a proxy (our own ideal cycle time); parts are moved by hand; Bosch time units are estimated by the Kaggle community, not published by Bosch [8].
- **Future research directions:** wireless boards, more stations and product types, and testing the WIP-limit alert on more real factory data.

RV College of Engineering

Timeline of the Project

Sprint	Dates	Work	Linked to	Status
1	23 to 29 Oct 2026	Problem, literature review, POs, line design, parts list, code skeleton	Phase I review (29 Oct)	**Done**
2	30 Oct to 1 Nov 2026	Wire all sensors, logger and dashboard running together	PO1, PO2	**In progress**
3	2 Nov to __	Count accuracy test, Bosch data mode, first runs	PO1, PO2, Phase II	Not started
4	__ to __	Baseline, rebalance and WIP-cap experiments (3 repeats each)	PO3, PO4	Not started
5	__ to __	Results, report, poster, PPT, paper draft, demo rehearsal	Phase III	Not started

Phase II date: __ / __ / 2026 **Phase III date: __ / __ / 2027** (dates to be announced; sprints will be fixed backwards from them)

RV College of Engineering

Conclusion

- **Summary of key points:** OptiProductio measures production count, cycle time, inventory and workflow on a 3-station model line, live, using an Arduino UNO, low-cost sensors and a Python dashboard.
- **What makes it different:** it adds a WIP-limit alert based on CONWIP [5], and it checks its own numbers against a hand tally and Little's law [7].
- **Final thought:** you cannot improve a line you cannot see. OptiProductio makes the line visible cheaply.
- **Next steps:** finish wiring by 1 Nov, run the count accuracy test (PO1), then bring partial results to Phase II.

RV College of Engineering

References

[1] Press Information Bureau. (2024, December 23). The MSME revolution: Transforming India's economic landscape. Government of India. https://static.pib.gov.in/WriteReadData/specificdocs/documents/2024/dec/doc20241223476301.pdf

[2] Hwang, G., Lee, J., Park, J., & Chang, T.-W. (2017). Developing performance measurement system for Internet of Things and smart factory environment. International Journal of Production Research, 55(9), 2590–2602. https://doi.org/10.1080/00207543.2016.1245883

[3] Ng Corrales, L. del C., Lambán, M. P., Hernandez Korner, M. E., & Royo, J. (2020). Overall equipment effectiveness: Systematic literature review and overview of different approaches. Applied Sciences, 10(18), 6469. https://doi.org/10.3390/app10186469

[4] Wicaksono, A. A., Ningsih, Y. K., & Surjati, I. (2024). Implementation of MQTT protocol on ESP32-based OEE analysis development board. Emitor: Jurnal Teknik Elektro, 24(2), 169–175. https://doi.org/10.23917/emitor.v24i2.3908

[5] Spearman, M. L., Woodruff, D. L., & Hopp, W. J. (1990). CONWIP: A pull alternative to kanban. International Journal of Production Research, 28(5), 879–894. https://doi.org/10.1080/00207549008942761

[6] Thürer, M., Li, S. S., Yang, C., Qu, T., & Huang, G. Q. (2023). Does regulating work-in-process increase throughput and reduce cycle times? An assessment by lab scale system models. In Advances in Production Management Systems (APMS 2023), IFIP AICT 691 (pp. 649–661). Springer. https://doi.org/10.1007/978-3-031-43670-3_45

[7] Little, J. D. C. (2011). Little's law as viewed on its 50th anniversary. Operations Research, 59(3), 536–549. https://doi.org/10.1287/opre.1110.0940

References (contd.)

[8] Mangal, A., & Kumar, N. (2017). Using big data to enhance the Bosch production line performance: A Kaggle challenge. arXiv. https://arxiv.org/abs/1701.00705

[9] Pettersen, J.-A., & Segerstedt, A. (2009). Restricted work-in-process: A study of differences between Kanban and CONWIP. International Journal of Production Economics, 118(1), 199–207.

[10] International Organization for Standardization. (2014). Automation systems and integration: Key performance indicators (KPIs) for manufacturing operations management. Part 2 (ISO 22400-2:2014). https://www.iso.org/standard/54497.html

[11] Cordon, C. (1995). Quality defaults and work-in-process inventory. European Journal of Operational Research, 80(2), 240–251.

[12] Bosch. (2016). Bosch production line performance [Data set]. Kaggle. https://www.kaggle.com/c/bosch-production-line-performance

[13] The Patents Act, 1970, § 3(k) (India). https://www.indiacode.nic.in/handle/123456789/1392

RV College of Engineering

Expected Outcome by the End of First Year

Product prototype: a working OptiProductio line and dashboard, released as open source (code, wiring table and build guide) so other labs can rebuild it.

Conference publication: a short paper in IEEE format on the prototype, the accuracy check and the WIP-limit experiments. Target venue to be chosen once results are in.

Research proposal: extend the work to more stations, wireless boards and more real factory data.

Patent: not planned. Software and algorithms on their own are generally not patentable in India (Patents Act, 1970, Section 3(k)) [13], so we will not claim one.


Thank You!